

Team My Big Fat Data

Project Name: Big Data Deduplicator

Members

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Project Description

This project facilitates the management and storage of big data for end-users through the use of innovative large scale data analysis algorithms as well as novel applications.

The Problem: The management of data, specially big data, introduces many problems users have to deal with because of the enormous amount of data that one must maintain in storage. In order to deal with this data we will need fast, efficient algorithms that can help us organize this data and present it in manner that is easier to understand. This is important because the usage and analysis of big data is crucial in many different job industries, therefore we need to create fast and efficient techniques to deal with this increasing amount of data.

Project Outcome

Document comparisons for deduplication in large scale databases is too slow when done sequentially. Our goal and vision for this project is to create an application using methods from machine learning and natural language processing to achieve efficient document deduplication. At the end of our project development and testing phase, we will have created a software application that can take large amounts of text documents as input and organize these files in a way that no “similar” or duplicate documents will be found in the resulting data.

Initial Project Milestones

- Learn to program in Scala
- Explore Spark and its built-in machine learning library
- Amazon Web Services will be used to provide us with the platform for Spark and Hadoop
- Explore other technologies such as ember, Node.js, and elastic
- Weekly visits at HG data's office for resources and meetings
- Assign roles within the group
- Visually represent the entire system (project requirements)

Courses of Action

Platform and Technologies: We will be using Spark and Hadoop as our tools of implementation for our data deduplication application. Amazon Web Services will provide the platform for the use of Spark and Hadoop. In addition, we will explore the usage of other software programming and design tools should they prove to be beneficial to our project.

Process Model: Bi-weekly sprint will be used to show current progress and assign new tasks for the upcoming weeks. Trello will be used to keep track of these sprint tasks and to show real-time progress for each teammate. Weekly meetings with TA and mentors will be used as resources for problems that we run into. Daily scrums will allow for communication over skype or google hangouts and will be used to talk about current progress or issues.